that axles be given a Magnaglo or Magnaflux test whenever a wheel is removed, regardless of length of service. The roller bearing inner races should also be removed so that the axle under the bearing race can be visually inspected for fretting and then Magnaflux tested. (See "Magnaflux inspection" at end of this instruction.)

Crack indications on an axle in the transverse plane are dangerous and the axle should be scrapped, unless individual railroad specifications authorize crack removal procedures. The axle should be scrapped if suspension bearing journals or the journals of axles for friction type boxes show evidence of bearing seizure accompanied by thermal cracks. Copper penetration makes the axle steel very brittle and subject to breakage. A distressed axle should never be reused until inspected for copper penetration and approved for use by the Railroad Engineer of Tests.

Axles should be checked for runout and should not exceed .030" total indicator reading with journal box bearing at zero. If the axle has over .030" total runout, it should be scrapped.

On switcher locomotives equipped with friction bearings, the journal taper should be checked. If taper exceeds 1/32" it should be reworked. On switcher locomotives equipped with roller bearings the taper should be reworked if it exceeds .001".

New and service dimensions for axles are shown in Fig. 4.

Axles and axle gears may be used until worn to maximum wear limits, if Magnaflux inspection reveals no defects.

If axles are not Magnafluxed at each wheel removal, it is recommended that the axle be scrapped after two pair of wheels have been worn to maximum wear limits.

In order to insure accurate axle measurements, the micrometers used should be checked against a known standard. Axles should always be measured when they are cold, never when they are warm from machining. All fillets and axle ends must be polished free of tool marks, and each time wheels are turned, all burrs or other irregularities should be polished off axle ends. If it is necessary to machine axle ends, the same amount should be removed from each end to insure that the axle is centered in the truck after assembly.

If the axle has a spline for a speed governor drive, the spline should be visually inspected and replaced if any of the teeth are found broken, chipped, cracked or have rounded edges.

The journals should also be checked for taper and machined if the taper exceeds 1/32" for the length of the journal. The micro inch finish of a maximum of 15 micro inches must be maintained.

If the wheel set has been removed from a truck involved in a wreck, the inner races must be removed and the journals of the axles should be Magnaglo or Magnaflux tested.

Axle Journal - Roller Bearing (See Fig. 4)

The minimum axle journal diameter is governed by the interference fit between the axle journal and the bearing inner race.

On axles manufactured prior to July, 1951, the new axle journal diameter was 6.503" to 6.5015". This provided a minimum interchangeable interference fit of .0015" to the inner race, which can have an I.D. of 6.500" to 6.499".

On axles manufactured after July, 1951, the new axle journal diameter is 6.504" to 6.503". This provides a minimum interference fit of .003" to the inner race.

A service limit of 6.5015" minimum axle journal diameter provides an interchangeable interference fit of .0015". If it is desired to use a selective fit, an axle journal diameter of 6.5005" can be used with an inner race having a minimum I.D. of 6.499" to obtain the .0015".